



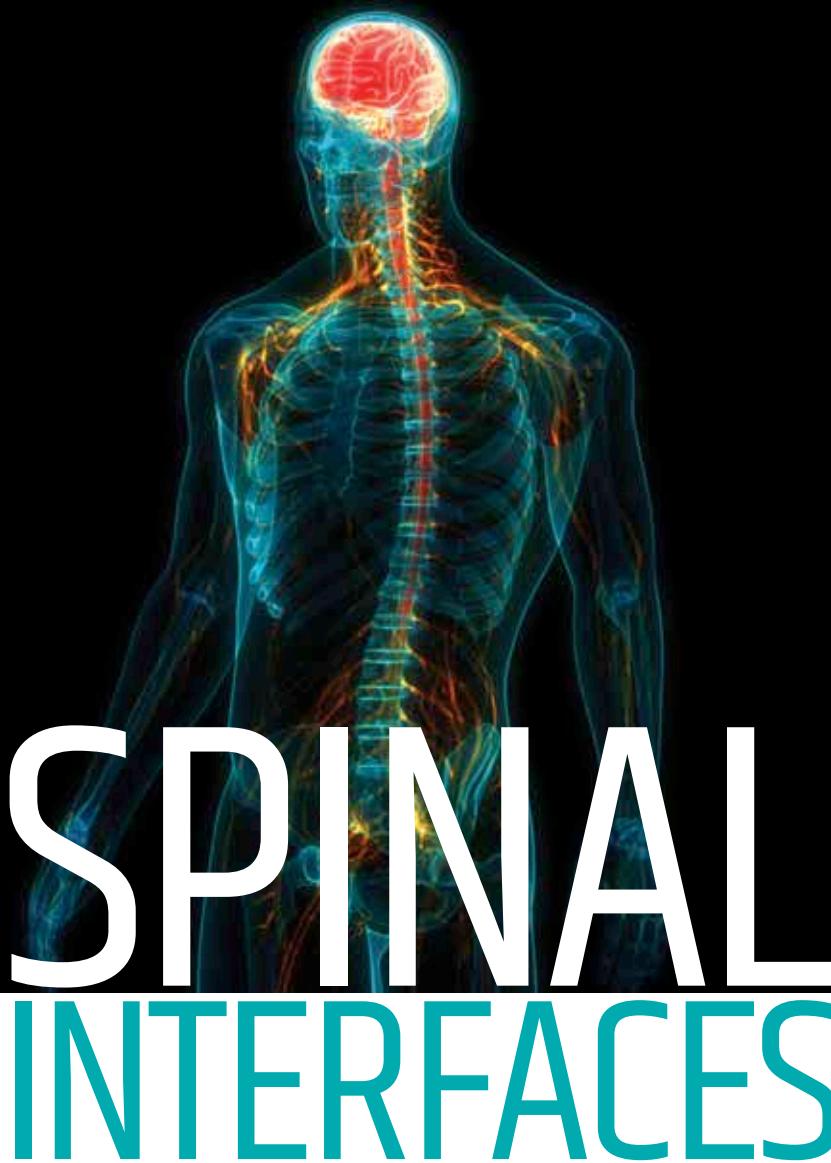
Breaching brain barriers

Scientists found a way to circumvent the brain's natural defences which obstruct the fighting of tumours. <https://digit.in/feb20-21>



Mummy voice recreated

Scientists 3D printed a 3000-year-old mummy's vocal tract and were able to hear what the mummy sounded like. <https://digit.in/feb20-22>



Links between computers and neurons need not be at the brain

Aditya Madanapalle | aditya@digit.in

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oused within the backbone, or the vertebral column is an extension of the brain, known as the spinal cord. At the neck, the spinal cord attaches to the brainstem, and is responsible for connecting most of

the nerves in the body to the brain and vice versa, although there are a few nerves in the head that originate directly from the brain. While the brain does all sorts of things, the spinal cord has very specific functions. It transmits the signals that control voluntary movements of the body, as well as relays back information from the sensing neurons. The spinal cord is also responsible for reflex actions.

Running through the centre of the backbone is a thin column known as the central or ependymal canal. This space is filled with cerebrospinal fluid, the same fluid that surrounds the brain in the skull. This is a delicate

part of the body, and any injuries can have serious consequences. Any kind of sudden, jerky movement, excessive pressure, or stretching apart of the vertebrae can permanently damage the spinal cord. Falls, motor vehicle accidents and sports-related injuries can also result in permanent damage. A whole range of inflammatory and infectious diseases including meningitis and tuberculosis can also result in spinal cord injuries. The resulting condition can be debilitating for anyone who experiences such injuries, with permanent paralysis, loss of sensory functions, loss of bladder control, and a dependence on medical apparatus for respiration. Depending on the particular vertebrae where the injury has occurred, there may be milder effects, such as paralysis in the wrists or hands, or an inability to grasp objects, and weak hand muscles. According to statistics by the World Health Organisation, every year between 250,000 to 500,000 people around the world suffer from spinal cord injuries.

After such an injury, an individual has to essentially withdraw from society. They have to depend on caregivers and assistive technologies, and basically live out the rest of their lives as outcasts. There are a number of researches being done to develop assistive technologies that can restore motor, and if possible sensory functions for these individuals, so that they can be reintegrated within society, and live a productive, mostly normal life. One of the new areas of research are spinal cord-computer interfaces, which are in theory simpler to realise than brain computer interfaces, because the spinal cord is relatively simpler than the brain. As the function of the spinal cord is to send the signals for movement, sensors tapping into it, can in theory derive more direct information on the intended types of movements, as compared to say an implant in the motor cortex of the brain, that is used for the same purpose, that is say transmitting signals to an exoskel-